

YrkesCo – Data Modeling Lab

Conceptual → Logical → Physical → Docker/Postgres validation

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github.com/JohnnyHyytiainen/lab_data_modeling_DE25_johnny_hyytiainen

Big picture

What this lab demonstrates

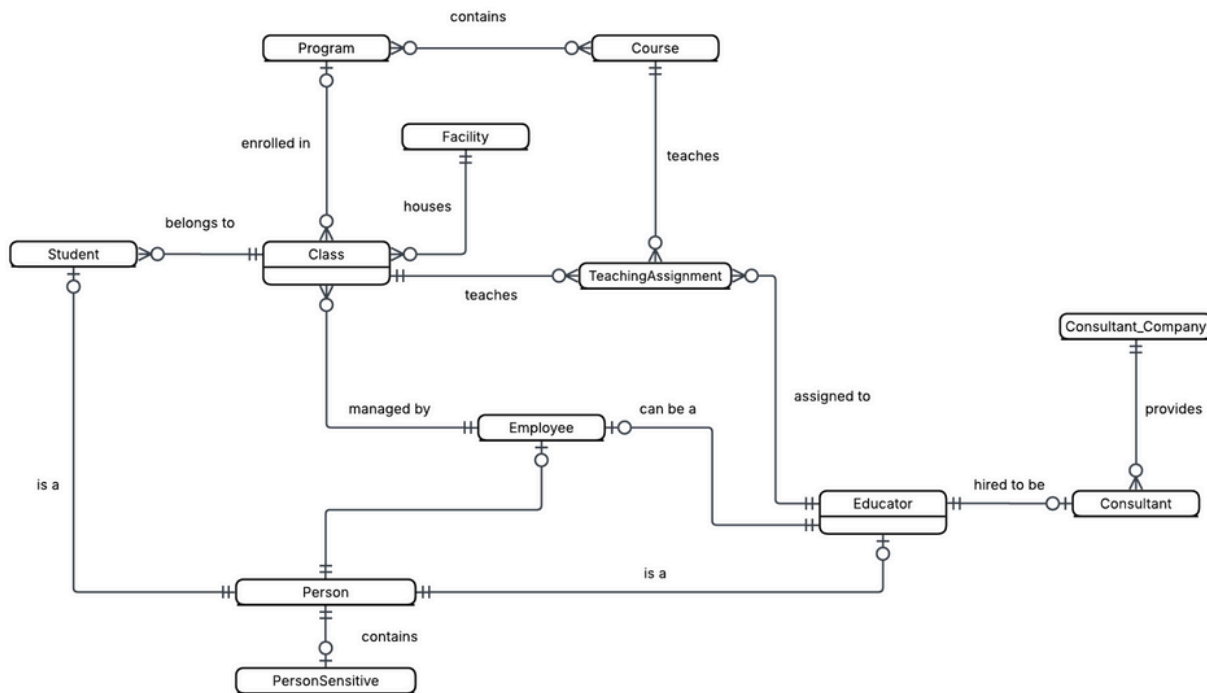
Requirements → **CDM** → **LDM** → **PDM** → **Docker/Postgres**

Key points

- Start from business requirements for a polytechnic school domain (YrkesCo)
- Model the domain in three layers: Conceptual (CDM) → Logical (LDM) → Physical (PDM)
- Implement the physical model in PostgreSQL, run in Docker, and validate via DDL/Seed/Queries/Sanity checks
- Keep evidence reproducible: scripts + screenshots + step-by-step output.md.

Conceptual model (CDM)

Entities + relationships (no data types yet)



What CDM captures

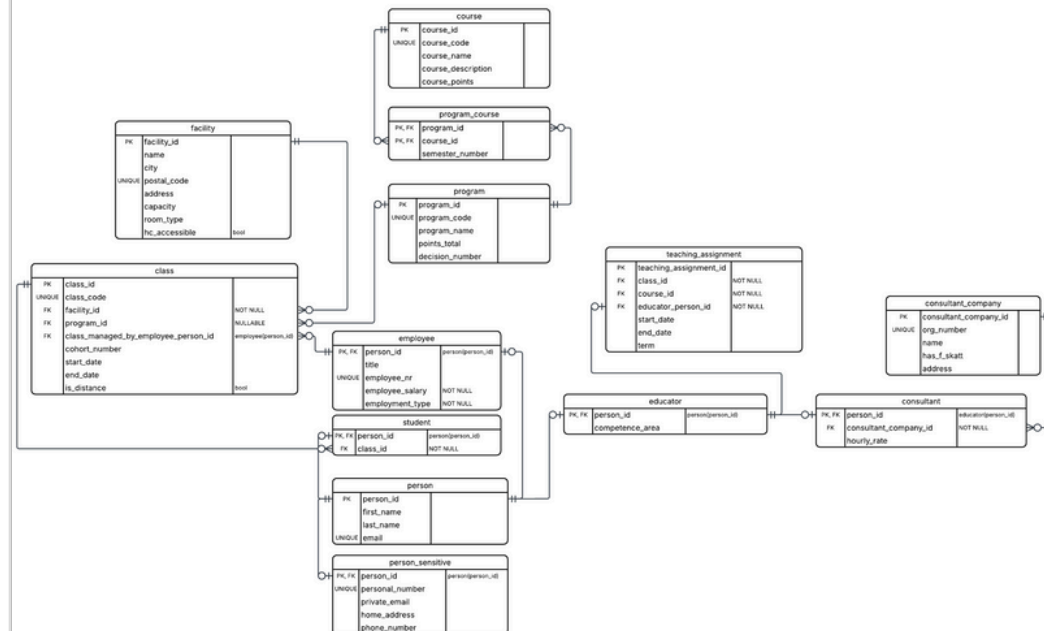
- Core things: Person, Student, Educator, Employee, Class, Program, Course, Facility, Consultant/Company, TeachingAssignment.
- Relationships first: who belongs to what, who teaches what, and where (Facility).
- Goal: agree on meaning with business language before thinking about columns and data types.

From CDM to LDM

Turning ideas into tables + keys

Main design choices

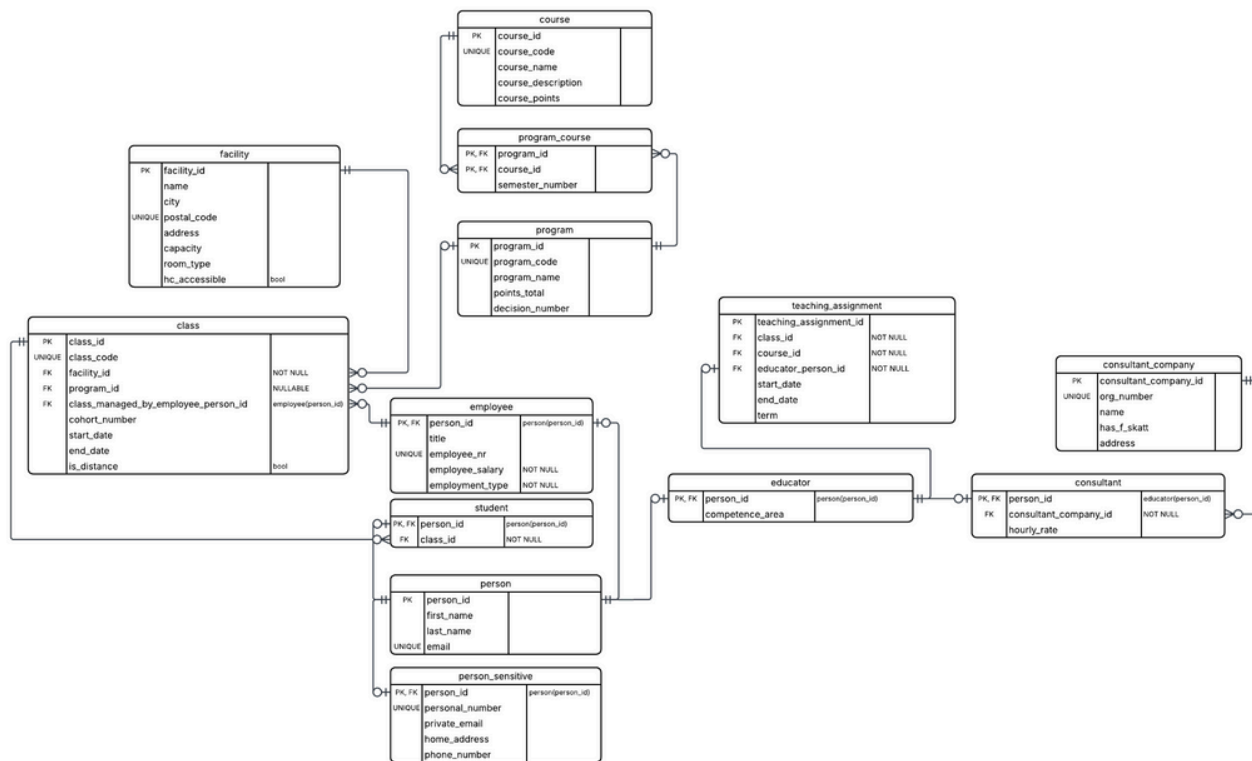
- Identify primary keys and foreign keys.
- Resolve many-to-many (Program <-> Course) with a junction table: program_course.
- Model inheritance with supertype/subtypes: Person as base, Student/Educator/Employee as subtypes (shared PK).
- Separate sensitive attributes into person_sensitive.



LDM

Logical model (LDM)

Tables, relationships and normalization

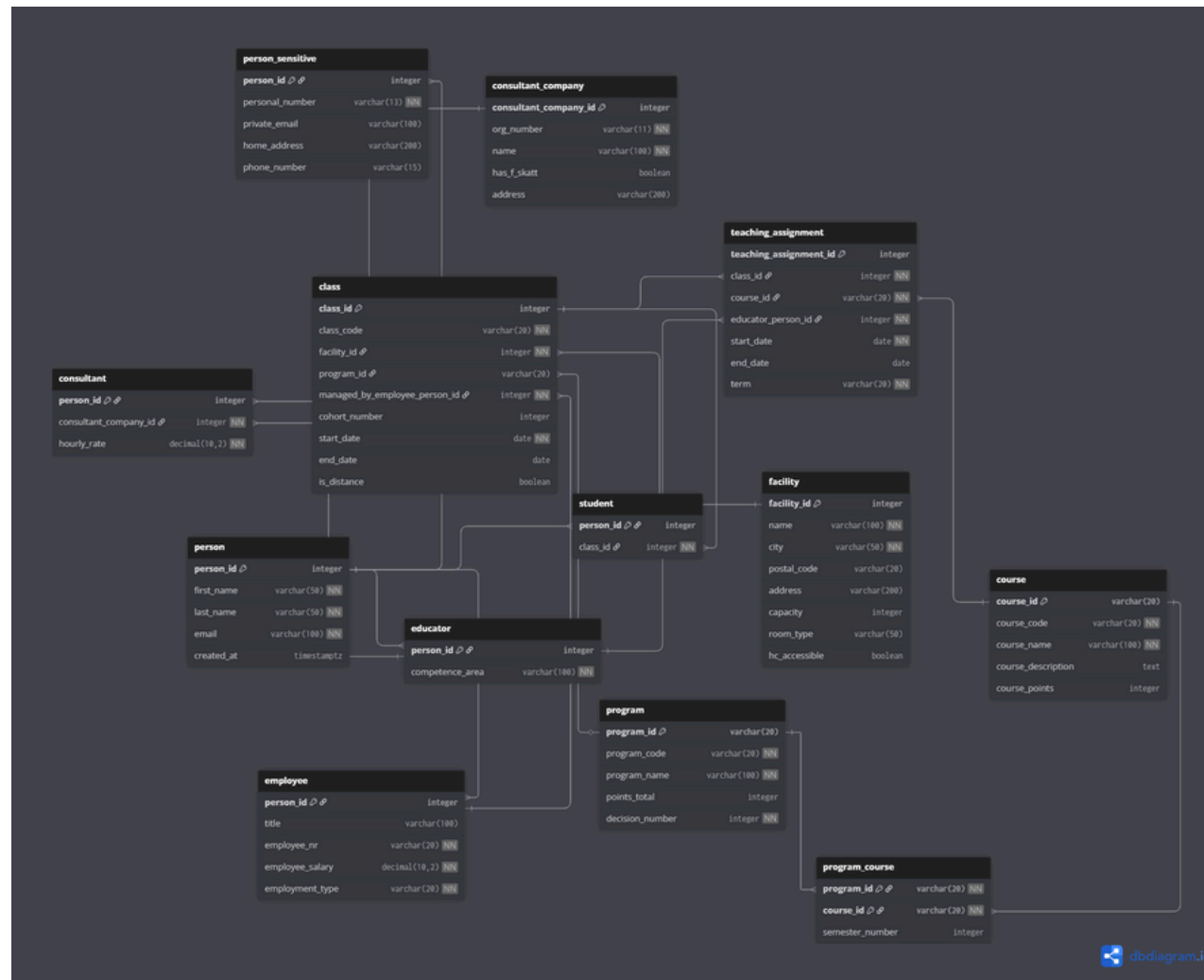


Why it looks like this

- 3NF mindset: keep attributes where they belong and avoid repeating groups.
- person_sensitive holds sensitive data separately (privacy + least access).
- teaching_assignment acts as a ternary relationship (Class + Course + Educator).
- Consultant is a specialization of Educator (consultant.person_id references educator.person_id).

From LDM to PDM

Datatypes, constraints and implementation details



Physical layer adds

- Concrete data types (varchar, integer, boolean, date, decimal).
- Constraints: NOT NULL, UNIQUE, CHECK patterns, FK constraints.
- Delete rules where needed (e.g. cascade for subtypes).
- Indexes/keys aligned with query paths (joins).

Docker + PostgreSQL setup

How the physical model is built and tested

```
$ docker compose -f docker/docker-compose.yml --env-file docker/.env up -d
[+] Running 3/3
 ✓ Network docker_default      Created      0.0s
 ✓ Volume docker_yrkesco_pgdata Created      0.0s
 ✓ Container lab_yrkesco_postgres Started       0.9s
```

Container up + healthy

Run steps (high level)

- dockercomposeup-d(uses docker/.env).
- 01_ddl.sql: rebuild schema (drop/create).
- 02_seed.sql: insert demo data (idempotent-ish via ON CONFLICT DO NOTHING).
- 03_queries.sql: read-only queries + \echo headings.
- 04_sanity_checks.sql: data quality & referential integrity checks.

Scripts execute with expected output

- More outputs are saved in docs/screenshots and documented in docs/evidence/output.md.

DDL

Seed

Queries (example)

Bug discovery & fix

Sanity checks caught a missing role assignment

feature	count	status
Fast anställda educators (BONUS)	0	MISSING

Bug detected

```
-- Eva = (fast anställd lärare) (educator + employee)
INSERT INTO employee (person_id, title, employee_nr, employee_salary, employment_type)
SELECT p.person_id, 'Utbildare', 'EMP-0002', 45000, 'tillsvidare'
FROM person p
WHERE p.email = 'eva.educator@yrkesco.se'
ON CONFLICT (person_id) DO NOTHING;
```

Fix applied in 02_seed.sql

What happened

- SanityCheck6flagged: 'Fast anställda educators' = 0 (missing).
- Root cause: Eva existed as person + educator, but not as employee.
- Fix: insert Eva into employee (shared PK pattern) and rerun seed.
- Validation: sanity checks pass; outputs documented with screenshots.

Wrap-up

What to look at in the repo

Repo highlights

- docs/: CDM/LDM/PDM diagrams + design notes.
- sql/: 01_ddl.sql, 02_seed.sql, 03_queries.sql, 04_sanity_checks.sql.
- docs/evidence/output.md: step-by-step reproduction + expected outputs.
- runbook_docker.md: practical command shortcuts and troubleshooting.